THE MARVELS OF RADIO-ACTIVITY.

(*Continued from page* 171)

of the leaves being indicative of the amount of Radium in a certain amount of sample.

Rutherford showed that the discharging effect is due to the production of *ions* or *charged particles* of the gas thru which the radiations pass. In an electric field, positive ions travel to the negative electrode and vice versa; thus causing the discharge of an electrified body. If a sufficiently strong field is used, the ions are all swept to the electrodes without appreciable loss. The rate of discharge then reaches a maximum, which is not altered by an increased voltage, this maximum current being called the saturation current. The ions produced are in every way identical with those produced by X-rays. This phenomenon of *ionization* is the basis for the conductivity of gases caused by radioactive substances.

Radium

Radium has been definitely determined as an element, atomic weight 226.2 (Mme. Curie). It imparts a brilliant red color to a flame and red predominates in its flame spectrum. The production of Radium is in this country largely from the ore carnotite, an Uranium oxid, found in Colorado and Utah. The Radium is extracted by chemical means from the ore together with Barium, from which it is separated by fractional crystallization. The bromid salt of Radium is slightly less soluble than that of Barium, so on cooling a solution, crystals richer in Radium than in Barium separate out first. After six or eight successive operations, pure crystals of Radium bromid are obtained, which are then ready for the market.

The prominence attained by Radium is due more to its ease of production and the amount of easily secured ore than to any exceptional properties it possesses over many of the other radioactive elements.

Besides the radiations given off by Radium, there is produced in addition a gas, known as "emanation." This gas is about a hundred thousand times as active as Radium itself. When introduced into a glass tube it causes a bright glow. This glow increases to a brilliant illumination when substances which phosphoresce, as zinc sulfid, are put into a tube containing emanation. Fig. 2.

By means of this emanation a finer estimation of amounts of Radium than by the ordinary electroscopic method can be made. The emanation is driven off by boiling and conducted into a suitable electroscope and the rate of collapsing of the leaves noted as compared to a standard. Quantities of radium as small as .000,000,000,001 gram can be detected and determined. This emanation method is used to determine the amount of Radium in rocks and minerals.

Another method for quantitative measurements of small amounts of Radium, when not less than 1/100 of a milligram is present, is to place the tube containing the Radium some distance from a lead screen and measure the rate of discharge of an electroscope, as compared with the rate caused by a standard amount of radium similarly placed. The material being investigated must be at least a month old, in order that the emanation be in equilibrium with the Radium (due to decay and recovery, as explained later). This method is simple and direct, as the tube need not be opened nor the material weighed.

The radioactive substance, Actinium, also gives off an emanation, whose activity dies in a few seconds. Polonium likewise undergoes a series of changes, there being considerable evidence that the final product is *lead*. Polonium is much more active than Radium, but occurs in smaller quantities. Marckwald obtained three milligrams from fifteen tons of pitchblende residue. The amount of Polonium in a Radium mineral is 1/5000 of the amount of the Radium.

Another body, known as "Ionium," has been recently separated with similar radioactive properties. Ionium compounds are several thousand times as active as those of Uranium. The especial interest in Ionium is that its decomposition product is Radium, altho its period of transformation is much longer than that of Radium.

It has been found that Uranium, Ionium, Radium, Actinium, Thorium—all break down, some giving off emanation, into new substances which in turn break down again into others, and so on thru the series. This phenomenon is nothing more or less than a spontaneous commutation of matter.

Perhaps the ancient Alchemists' ideas were not entirely wrong!

(*To Be Continued*)

NIKOLA TESLA RECEIVES THE "EDISON MEDAL."

Nikola Tesla, the famous electrical wizard, who was awarded the seventh Edison medal on December 13, 1916, "for meritorious achievements in his early original work in polyphase and high-frequency electric currents," received the medal at a presentation made at the annual meeting of the American Institute of Electrical Engineers, on May 18.

The Edison Medal was established upon the initiative of a group of friends and associates of Thomas A. Edison, for the purpose of recounting and celebrating the achievements of a quarter of a century in the art of electric lighting, with which the name of Edison is imperishably identified. It was decided that the most effective means of accomplishing this object would be by the establishment of a gold medal, which should, during the centuries to come, serve as an honorable incentive to scientists, engineers and artisans to maintain by their works a high standard of accomplishment.

The Edison Medal was, therefore, established and endowed with a trust fund, under an indenture dated February 11, 1904, whereby the American Institute of Electrical Engineers agreed to award the medal annually. It is awarded each year by a committee consisting of 24 members of the Institute to a resident of the United States of America and its dependencies, or of the Dominion of Canada, "for meritorious achievement in electrical science or electrical engineering or the electrical arts."

B. A. Behrend said:—

"By an extraordinary coincidence, it is exactly twenty-nine years ago, to the very day and hour, that there stood before this Institute Nikola Tesla, and he read a description of his great discovery of the generation and utilization of polyphase alternating currents. He left nothing to be done for those who followed him. His paper contained the skeleton even of the mathematical theory.

"Three years later, in 1891, there was given the first great demonstration, by Swiss engineers, of the transmission of power at 30,000 volts from Lauffen to Frankfort by means of Mr. Tesla's system. A few years later this was followed by the development of the Cataract Construction Company, under the presidency of our member, Edward D. Adams, and with the aid of the engineers of the Westinghouse Company. It is interesting to recall here to-night that in Lord Kelvin's report to Mr. Adams, Lord Kelvin recommended the use of direct current for the development of power at Niagara Falls and for its transmission to Buffalo.

Nikola Tesla, Prince of Electrical Inventors, Who Was Recently Awarded the "Edison Medal."

"The basis for the theory of the operating characteristics of Mr. Tesla's rotating-field induction motor, so necessary to its practical development, was laid by the brilliant French savant Prof. André Blondel, and by Professor Kapp of Birmingham. It fell to my lot to complete their work and to co-ordinate—by means of the simple 'circle diagram'—the somewhat mysterious and complex experimental phenomena. As this was done twenty-one years ago, it is particularly pleasing to me, upon the coming of age of this now universally accepted theory—tried out by application to several million horse-power of machines operating in our great industries—to pay my tribute to the inventor of the motor and the system which have made possible the electric transmission of energy. His name marks an epoch in the advance of electrical science. From that work has sprung a revolution in the electrical art."

ELECTRIC RESISTANCE OF SELENIUM CELLS.

According to Professor H. Greinacher of Zurich, selenium cells of the original Shalford Bidwell type, which he studied together with Mr. C. W. Miller, behave with respect to alternating currents as they behave when exposed to light, and show polarity when traversed by continuous currents. Communicating his observations to the German Physical Society, Greinacher stated that the resistance of the selenium cell rose when direct current flowed in the dark, that this increase in resistance was different for positive and for negative currents and increased with the time, and that selenium cells acted in a certain sense like current rectifiers. These statements are questioned by Dr. Robert Furstenau of Berlin, who, in experimenting with hundreds of selenium cells with similar bridge arrangements as Greinacher, had never noticed any of these effects. Furstenau suggests that Greinacher's cells had been of peculiar kind. That selenium cells are very sensitive to moisture, and that the electrode material may have peculiar puzzling effects, is fairly well understood, and these features may explain some of the controversial statements made from time to time.